authorizationTokenManagement

Service Description

Service ID: *"authorizationTokenManagement"*

Abstract
This document describes the `authorizationTokenManagement` service, version 1.0.0, produced by the ConsumerAuthorization core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `authorizationTokenManagement` service, which provides operator management of authorization tokens and of provider encryption keys. It is produced by the ConsumerAuthorization system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

New in Arrowhead 5.

2.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration calls `generate-tokens` on behalf of consumers during orchestration (it is in `unbounded.token.generation.whitelist`). Operators query and revoke tokens and register or remove encryption keys of providers.

2.3 Important Delimitations and Dependencies

Unbound token generation (bypassing the authorization check) is only allowed for systems in `unbounded.token.generation.whitelist`.

3 Service Interface

This section describes the interfaces to the `authorizationTokenManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

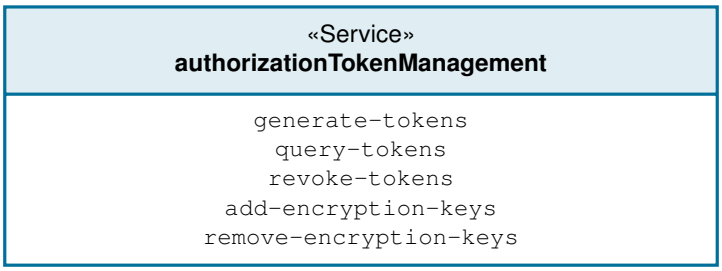


Figure 1: SysML-style block of the `authorizationTokenManagement` service and its operations.

The following interface operations are available.

3.1 operation generate-tokens (AuthorizationTokenGenerationMgmtListRequest) : AuthorizationTokenMgmtListResponse

Generates tokens on behalf of consumers. With `unbound=true` the authorization check is skipped (only for systems in `unbounded.token.generation.whitelist`).

Optional parameter: `unbound`.

3.2 operation query-tokens (AuthorizationTokenQueryRequest) : AuthorizationTokenMgmtListResponse

Returns a page of tokens matching the filter.

3.3 operation revoke-tokens (List<Identifier>)

Revokes tokens by token reference.

3.4 operation add-encryption-keys (AuthorizationMgmtEncryptionKeyRegistrationListRequest) : AuthorizationMgmtEncryptionKeyListResponse

Registers encryption keys on behalf of providers.

3.5 operation remove-encryption-keys (List<Name>)

Removes the encryption keys of the named providers.

4 Information Model

Here, all data objects that can be part of the `authorizationTokenManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.11, which are used to represent things like hashes and identifiers.

4.1 struct AuthorizationMgmtEncryptionKeyListResponse

List of registered encryption keys.

Field	Type	Description
entries	List<AuthorizationMgmtEncryptionKeyResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.2 struct AuthorizationMgmtEncryptionKeyRegistrationListRequest

Encryption keys to register on behalf of providers.

Field	Type	Description
list	List<AuthorizationMgmtEncryptionKeyRegistrationRequest>	The list of request items.

4.3 struct AuthorizationMgmtEncryptionKeyRegistrationRequest

Encryption key of one provider.

Field	Type	Description
systemName	Name	Provider system the key belongs to.
key	String	The encryption key, as a string.
algorithm	String	Encryption algorithm of the key, e.g. `AES/ECB/PKCS5Padding` or `AES/CBC/PKCS5Padding`.

4.4 struct AuthorizationMgmtEncryptionKeyResponse

A registered encryption key.

Field	Type	Description
systemName	Name	Name of the system, PascalCase (e.g. `TemperatureSensor`).
rawKey	String	The registered encryption key.
algorithm	String	Encryption algorithm of the key, e.g. `AES/ECB/PKCS5Padding` or `AES/CBC/PKCS5Padding`.
keyAdditive	String	Initialization vector (CBC mode), Base64; empty for ECB.
createdAt	DateTime	UTC time of creation.

4.5 struct AuthorizationTokenGenerationMgmtListRequest

Tokens to generate on behalf of consumers.

Field	Type	Description
list	List<AuthorizationTokenGenerationMgmtRequest>	The list of request items.

4.6 struct AuthorizationTokenGenerationMgmtRequest

One token to generate on behalf of a consumer.

Field	Type	Description
tokenVariant	String	Requested token variant, as interface security policy (e.g. `TIME_LIMITED_TOKEN_AUTH`, `RSA_SHA512_JSON_WEB_TOKEN_AUTH`).
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
consumerCloud	Name	Cloud identifier of the consumer system.
consumer	Name	Name of the consumer system.
provider	Name	Name of the provider system.
target	Name	Target of the policy or token: a service definition or event type.
scope	Name	Operation the token or policy is limited to; empty for all operations.
expiresAt	DateTime	Requested expiry; default `token.time.limit` from now.
usageLimit	Number	Requested uses; default `simple.token.usage.limit`.

4.7 struct AuthorizationTokenMgmtListResponse

List of generated tokens.

Field	Type	Description
entries	List<AuthorizationTokenResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.8 struct AuthorizationTokenQueryRequest

Filter for generated tokens.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
requester	Name	Name of the system that requested the token.
tokenType	String	Token type to match.
consumerCloud	Name	Cloud identifier of the consumer system.
consumer	Name	Name of the consumer system.
provider	Name	Name of the provider system.
targetType	String	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.

4.9 struct AuthorizationTokenResponse

A generated token with its bookkeeping data.

Field	Type	Description
tokenType	AuthorizationTokenType	Token type: `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN` or `TRANSLATION_BRIDGE_TOKEN`.
variant	String	Token variant (interface security policy).
token	Token	The token, for simple tokens only in hashed form.
tokenReference	Identifier	Reference of the token, used to revoke it.
requester	Name	Name of the system that requested the token.
consumerCloud	Name	Cloud identifier of the consumer system.
consumer	Name	Name of the consumer system.
provider	Name	Name of the provider system.
targetType	AuthorizationTargetType	Target type: `SERVICE_DEF` or `EVENT_TYPE`.
target	Name	Target of the policy or token: a service definition or event type.
scope	Name	Operation the token or policy is limited to; empty for all operations.
createdAt	DateTime	UTC time of creation.
usageLimit	Number	Number of allowed uses (usage-limited tokens).
usageLeft	Number	Remaining uses of a usage-limited token.
expiresAt	DateTime	UTC expiry time; no value means no expiry.

4.10 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

4.11 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
AuthorizationTargetType	One of `SERVICE_DEF`, `EVENT_TYPE`.
AuthorizationTokenType	One of `TIME_LIMITED_TOKEN`, `USAGE_LIMITED_TOKEN`, `SELF_CONTAINED_TOKEN`, `TRANSLATION_BRIDGE_TOKEN`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.
Token	An opaque string used as credential.

5 References

[1] J. Delsing, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, consumerauthorization/documentation/v5.3.0/ConsumerAuthorization-v5.3.0-SysD.pdf.

[2] ——, “authorizationTokenManagement HTTP/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationTokenManagement-HTTP-IDD.pdf.

[3] ——, “authorizationTokenManagement MQTT/TLS/JSON – Interface Design Description (ConsumerAuthorization),” Eclipse Arrowhead, IDD 5.3.0, 2026, consumerAuthorization-v5.3.0-authorizationTokenManagement-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	